A TRANSFORMATION SPLITTING THE AUGMENTATION GIVES A SECTION OF THE AUGMENTED GODEMENT CONSTRUCTION

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-godement-section (source lines 6573-6584); the proof environment begins on the second line after it and is its own proof, closing at source line 6609.

Editorial note. The statement invokes one author example of the same chapter by label; it is retained as a context span and the generator expands the context transitively (budget 800 lines for this card), while reference targets in other chapters that carry canonical project tags are auto-declared as non-load-bearing source pointers.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: a transformation splitting the augmentation gives a section of the augmented Godement construction.

Allowed standard prerequisites (editorial).

- The Godement construction setting. The functors F, G, the simplicial object X with its augmentation epsilon : X to Y and the maps d and s, as fixed in the author's Example example-godement-functorial (retained as context).. It is the setting of the statement and fixes the notation (source lines 6575-6583).
- Whiskering of natural transformations. Horizontal composition of natural transformations, written star, and its associativity and compatibility with identities.. It is what defines h_n and what makes h a morphism of simplicial objects (source lines 6589-6597).
- Simplicial objects and the maps X_0 to X_n. For a simplicial object X the maps u_n : X_0 to X_n induced by the unique morphism [n] to [0] assemble into a morphism from the constant simplicial object on X_0 to X.. It shows that the family (h_n) is a morphism of simplicial objects (source lines 6593-6597).

Source reference. Original lemma 003D: sources/stacks-chow/tags.txt, source lines 130–130. Cross-chapter pointer whose target carries the canonical tag 003D in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 003F: sources/stacks-chow/tags.txt, source lines 132–132. Cross-chapter pointer whose target carries the canonical tag 003F in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Example 0.1. Let $\mathcal{C}$, Y, d, s be as in Example 0.2 satisfying the equations of Lemma 0.3. Given functors $F : \mathcal{A} \to \mathcal{C}$ and $G : \mathcal{C} \to \mathcal{B}$ we obtain a simplicial object $G \circ X \circ F$ in the category of functors from $\mathcal{A}$ to $\mathcal{B}$ which comes with an augmentation to $G \circ F$.

Example 0.2. Let $Y : \mathcal{C} \to \mathcal{C}$ be a functor from a category to itself and suppose given transformations of functors

$$d : Y \longrightarrow \mathrm{id}_{\mathcal{C}} \quad \text{and} \quad s : Y \longrightarrow Y \circ Y$$

Using these transformations we can construct something that looks like a simplicial object. Namely, for $n \geq 0$ we define

$$X_n = Y \circ \ldots \circ Y \quad (n+1 \text{ compositions})$$

Observe that $X_{n+m+1} = X_n \circ X_m$ for $n, m \geq 0$. Next, for $n \geq 0$ and $0 \leq j \leq n$ we define using notation as in Categories, Section 003D

$$d_j^n = 1_{X_{j-1}} \star d \star 1_{X_{n-j-1}} : X_n \to X_{n-1} \quad \text{and} \quad s_j^n = 1_{X_{j-1}} \star s \star 1_{X_{n-j-1}} : X_n \to X_{n+1}$$

So d_j^n, resp. s_j^n is the natural transformation using d, resp. s on the jth Y (counted from the left) in the composition defining X_n.

Lemma 0.3. *In Example 0.2 if*

$$1_Y = (d \star 1_Y) \circ s = (1_Y \star d) \circ s \quad \textit{and} \quad (s \star 1) \circ s = (1 \star s) \circ s$$

then $X = (X_n, d_j^n, s_j^n)$ is a simplicial object in the category of endofunctors of $\mathcal{C}$ and $d : X_0 = Y \to id_{\mathcal{C}}$ defines an augmentation.

Proof. To see that we obtain a simplicial object we have to check that the relations (1)(a) – (e) of Lemma 0.4 are satisfied. We will use the short hand notation

$$1_a = 1_{X_{a-1}} = 1_Y \star \ldots \star 1_Y \quad (a \text{ factors})$$

for $a \geq 0$. With this notation we have

$$d_j^n = 1_j \star d \star 1_{n-j} \quad \text{and} \quad s_j^n = 1_j \star s \star 1_{n-j}$$

We are repeatedly going to use the rule that for transformations of funtors a, a', b, b' we have $(a' \circ a) \star (b' \circ b) = (a' \star b') \circ (a \star b)$ provided that the $\star$ and $\circ$ compositions in this formula make sense, see Categories, Lemma 003F.

Condition (1)(a) always holds (no conditions needed on d and s). Namely, let $0 \leq i < j \leq n+1$. We have to show that $d_i^n \circ d_j^{n+1} = d_{j-1}^n \circ d_i^{n+1}$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star d \star 1_{n+1-j}) = (1_{j-1} \star d \star 1_{n+1-j}) \circ (1_i \star d \star 1_{n+1-i})$$

We can rewrite the left hand side as

$$\begin{aligned} & (1_i \star d \star 1_{j-i-1} \star 1_{n+1-j}) \circ (1_i \star 1_1 \star 1_{j-i-1} \star d \star 1_{n+1-j}) \\ & = 1_i \star ((d \star 1_{j-i-1}) \circ (1_1 \star 1_{j-i-1} \star d)) \star 1_{n+1-j} \\ & = 1_i \star d \star 1_{j-i-1} \star d \star 1_{n+1-j} \end{aligned}$$

The second equality is true because $d \circ 1_1 = d$ and $1_{j-i} \circ (1_{j-i-1} \star d) = 1_{j-i-1} \star d$. A similar computation gives the same result for the right hand side.

We check condition (1)(b). Let $0 \leq i < j \leq n-1$. We have to show that $d_i^n \circ s_j^{n-1} = s_{j-1}^{n-2} \circ d_i^{n-1}$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j-1} \star s \star 1_{n-1-j}) \circ (1_i \star d \star 1_{n-1-i})$$

By the same kind of calculus as in case (1)(a) both sides simplify to $1_i \star d \star 1_{j-i-1} \star s \star 1_{n-j-1}$.

We check condition (1)(c). Let $0 \leq j \leq n-1$. We have to show $\mathrm{id} = d_j^n \circ s_j^{n-1} = d_{j+1}^n \circ s_j^{n-1}$, i.e.,

$$1_n = (1_j \star d \star 1_{n-j}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j+1} \star d \star 1_{n-j-1}) \circ (1_j \star s \star 1_{n-1-j})$$

This is easily seen to be implied by the first assumption of the lemma.

We check condition (1)(d). Let $0 < j+1 < i \leq n$. We have to show $d_i^n \circ s_j^{n-1} = s_j^{n-2} \circ d_{i-1}^{n-1}$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_j \star s \star 1_{n-2-j}) \circ (1_{i-1} \star d \star 1_{n-i})$$

By the same kind of calculus as in case (1)(a) both sides simplify to $1_j \star s \star 1_{i-j-2} \star d \star 1_{n-i}$.

We check condition (1)(e). Let $0 \leq i \leq j \leq n-1$. We have to show that $s_i^n \circ s_j^{n-1} = s_{j+1}^n \circ s_i^{n-1}$, i.e.,

$$(1_i \star s \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j+1} \star s \star 1_{n-1-j}) \circ (1_i \star s \star 1_{n-1-i})$$

By the same kind of calculus as in case (1)(a) this reduces to

$$(s \star 1_{j-i+1}) \circ (1_{j-i} \star s) = (1_{j-i+1} \star s) \circ (s \star 1_{j-i})$$

If $j = i$ this is exactly one of the two assumptions of the lemma. For $j > i$ left and right hand side both reduce to the equality $s \star 1_{j-i-1} \star s$ by calculations similar to those we did in case (1)(a).

Finally, in order to show that d defines an augmentation we have to show that $d \circ (1_1 \star d) = d \circ (d \star 1_1)$ which is true because both sides are equal to $d \star d$. □

Lemma 0.4. *Let $\mathcal{C}$ be a category.*

1. *Given a simplicial object U in $\mathcal{C}$ we obtain a sequence of objects $U_n = U([n])$ endowed with the morphisms $d_j^n = U(\delta_j^n) : U_n \to U_{n-1}$ and $s_j^n = U(\sigma_j^n) : U_n \to U_{n+1}$. These morphisms satisfy the opposites of the relations displayed in Lemma 0.6, namely*
   (a) *If $0 \leq i < j \leq n+1$, then $d_i^n \circ d_j^{n+1} = d_{j-1}^n \circ d_i^{n+1}$.*
   (b) *If $0 \leq i < j \leq n-1$, then $d_i^n \circ s_j^{n-1} = s_{j-1}^{n-2} \circ d_i^{n-1}$.*
   (c) *If $0 \leq j \leq n-1$, then $\mathrm{id} = d_j^n \circ s_j^{n-1} = d_{j+1}^n \circ s_j^{n-1}$.*
   (d) *If $0 < j+1 < i \leq n$, then $d_i^n \circ s_j^{n-1} = s_j^{n-2} \circ d_{i-1}^{n-1}$.*
   (e) *If $0 \leq i \leq j \leq n-1$, then $s_i^n \circ s_j^{n-1} = s_{j+1}^n \circ s_i^{n-1}$.*
2. *Conversely, given a sequence of objects U_n and morphisms d_j^n, s_j^n satisfying (1)(a) – (e) there exists a unique simplicial object U in $\mathcal{C}$ such that $U_n = U([n])$, $d_j^n = U(\delta_j^n)$, and $s_j^n = U(\sigma_j^n)$.*
3. *A morphism between simplicial objects U and U' is given by a family of morphisms $U_n \to U'_n$ commuting with the morphisms d_j^n and s_j^n.*

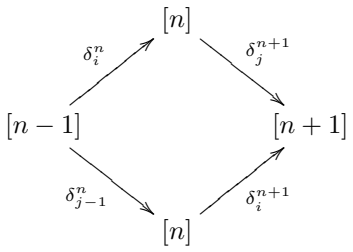
Proof. This follows from Lemma 0.5. □

Lemma 0.5. *The category Δ is the universal category with objects $[n]$, $n \geq 0$ and morphisms δ^n_j and σ^n_j such that (a) every morphism is a composition of these morphisms, (b) the relations listed in Lemma 0.6 are satisfied, and (c) any relation among the morphisms is a consequence of those relations.*

Proof. Omitted. □

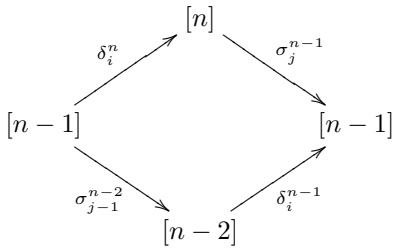
Lemma 0.6. *The morphisms δ^n_j and σ^n_j satisfy the following relations.*

(1) *If $0 \leq i < j \leq n+1$, then $\delta^{n+1}_j \circ \delta^n_i = \delta^{n+1}_i \circ \delta^n_{j-1}$. In other words the diagram*



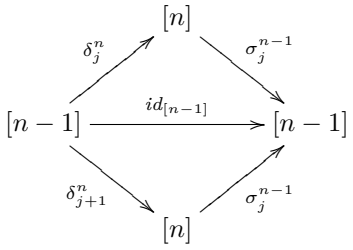
commutes.

(2) *If $0 \leq i < j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_i = \delta^{n-1}_i \circ \sigma^{n-2}_{j-1}$. In other words the diagram*



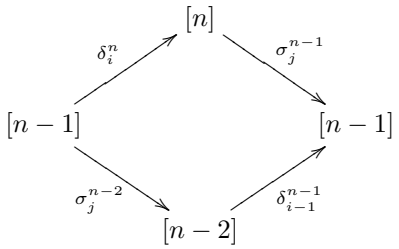
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(3) *If $0 \leq j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_j = id_{[n-1]}$ and $\sigma^{n-1}_j \circ \delta^n_{j+1} = id_{[n-1]}$. In other words the diagram*



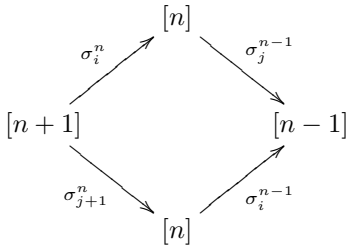
commutes.

(4) *If* $0 < j+1 < i \leq n$, *then* $\sigma_j^{n-1} \circ \delta_i^n = \delta_{i-1}^{n-1} \circ \sigma_j^{n-2}$. *In other words the diagram*



commutes.

(5) *If* $0 \leq i \leq j \leq n-1$, *then* $\sigma_j^{n-1} \circ \sigma_i^n = \sigma_i^{n-1} \circ \sigma_{j+1}^n$. *In other words the diagram*



commutes.

Proof. Omitted. □

Lemma 0.7. *Let* $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, Y, d, s, F, G *be as in Example 0.1. Given a transformation of functors* $h_0 : G \circ F \to G \circ Y \circ F$ *such that*

$$1_{G \circ F} = (1_G \star d \star 1_F) \circ h_0$$

Then there is a morphism $h : G \circ F \to G \circ X \circ F$ *of simplicial objects such that* $\epsilon \circ h = id$ *where* $\epsilon : G \circ X \circ F \to G \circ F$ *is the augmentation.*

Proof. Denote $u_n : Y = X_0 \to X_n$ the map of the simplicial object X corresponding to the unique morphism $[n] \to [0]$ in Δ. Set $h_n : G \circ F \to G \circ X_n \circ F$ equal to $(1_G \star u_n \star 1_F) \circ h_0$.

For any simplicial object $X = (X_n)$ in any category $u = (u_n) : X_0 \to X$ is a morphism from the constant simplicial object on X_0 to X. Hence h is a morphism of simplicial objects because it is the composition of $1_G \star u \star 1_F$ and h_0.

Let us check that $\epsilon \circ h = \mathrm{id}$. We compute

$$\epsilon_n \circ (1_G \star u_n \star 1_F) \circ h_0 = \epsilon_0 \circ h_0 = \mathrm{id}$$

The first equality because ϵ is a morphism of simplicial objects and the second equality because $\epsilon_0 = (1_G \star d \star 1_F)$ and we can apply the assumption in the statement of the lemma. □